Ioana Sarah Aizic

Final Honours Research Project Report

1. Introduction

The basis of my Honour's Research Project lies in constraint-based metabolic modelling, in which an algorithm maps gene expression data onto the Human1 Genome Scale Model to identify the most "optimal" pathway via a penalty minimisation calculation. This *in silico* modelling approach allows for the effects of medications to be modelled, providing a deeper understanding of their metabolic and toxicity pathways without causing harm to humans or animals. This project has focused on mapping the pathways of paracetamol degradation and metabolic reactions affected by paracetamol toxicity.

Paracetamol, or acetaminophen (APAP), is widely regarded as a "cure-all" medication (*Acute Liver Failure*, n.d.), but it is exactly its overuse and lack of awareness about correct dosages that make it the leading cause of liver failure in the US (Agrawal et al., 2025). At toxic doses, APAP is metabolised into high amounts of N-acetyl-p-benzoquinone imine (NAPQI) (Agrawal et al., 2025), which depletes the liver's glutathione (GSH) levels. GSH acts as an antioxidant and protects liver cells from oxidative damage (Forman et al., 2009), so when its concentrations are decreased, NAPQI damages proteins in the liver, eventually leading to liver failure. The pathways leading to the formation of NAPQI are well-researched; however, its neutralisation mechanism using the antioxidant GSH is not well understood. Furthermore, the depletion of GSH at toxic doses of paracetamol has severe effects on the normal metabolic functioning of hepatocytes, and the exact mechanism of damage is yet to be fully figured out.

In order to understand the logic of the MaCSBio pathway optimisation algorithm and how Python can be used to parse the data output by it, in September, I wrote pieces of code meant to analyse the reactions taken in different samples. I displayed this in the form of co-occurrence and row clustering matrices. These are meant to show core (always present) and variable (only present in some samples) reactions, as well as whether a reaction was present or absent in a specific sample. These matrices were used by an MSP project group at MaCSBio for further analysis.

After this, I started searching for gene expression information from hepatocytes at different doses of paracetamol. Before starting my Student-Led Project (SLP) in June 2025, I worked on parsing the dataset and normalising the data so that it could be used to map the relevant reactions. The project focused on updating the Human1 model to include the correct metabolites, reactions, and gene rules needed to map the three possible pathways of acetaminophen degradation. This was done in order to understand which of these three pathways would be chosen in healthy vs in overdosed samples.

Using the findings and further improvements from this project, I optimised the degradation task, model, and the expression dataset. While the SLP did yield promising findings for this task, after running the algorithm on the biologically correct reactions, it became clear that the idea of using flexible tasks to analyse this led to incorrect conclusions and cannot be done using this algorithm. Because of this, the flexible task was split into three different tasks and analysed separately (Section 2.1).

GSH de novo synthesis was one of the tasks which was expanded during the SLP, which is why it was further analysed using the updated expression dataset (Section 2.1). Metabolic reactions not related to APAP degradation and how they are affected by this drug's toxic effects were also examined. Thus, the citric acid cycle (Section 2.3) and arachidonic acid metabolic pathways (Section 2.4) were mapped and analysed.

2. Results and discussion

2.1. Paracetamol Degradation

Paracetamol is known to degrade within the body through one of three pathways: sulphation, glucuronidation, or NAPQI formation. A flexible task was made to analyse which pathway would be chosen based on the time-frame of paracetamol treatment.

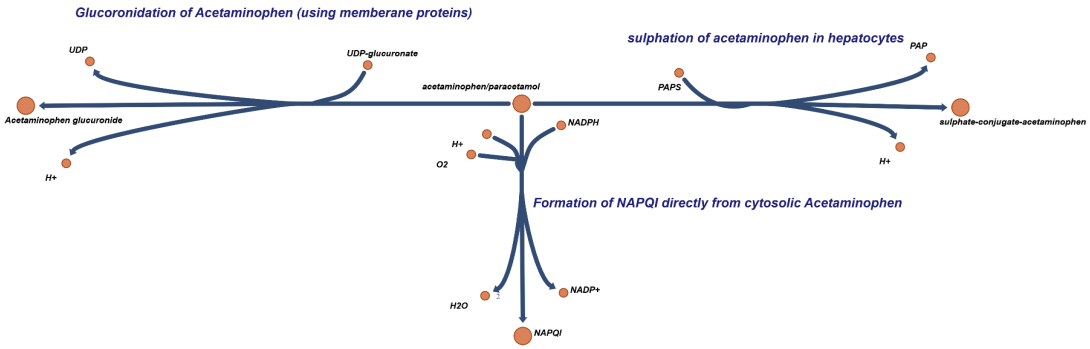


Figure 1. *Degradation pathways of paracetamol*

Following literature research, the pathways present in the model were updated to include the biologically correct enzymes and compartments and the results can be seen in Figure 1.

The flexible task failed to provide any useful information about the pathways preferred in each sample, since the expression values of the enzymes were not comparable. The glucuronosyltransferase family of enzymes, which facilitates glucuronidation, had significantly

higher expression than the cytochrome P450 family or sulfotransferase family, which catalyse NAPQI formation and sulphation, respectively. This made it such that the glucuronidation pathway was always preferred, even though it is not biologically feasible. To allow for some analysis to be done on this task, the three pathways were separated into different tasks and their expression values were analysed individually.

A general trend (Appendix A.) can be observed by looking at the increase or decrease in expression values. For sample 2 (30 minutes of treatment with paracetamol), the expression of the sulphation pathway is increased compared to the healthy sample. Based on these results, it could be interpreted that at the beginning of paracetamol ingestion, the sulphation pathway is chosen, and after the therapeutic dose is reached, this pathway becomes saturated (Li et al., 2021), which is biologically correct.

Until 12 hours of treatment, the enzyme expression of this pathway is decreased, while glucuronidation is increasing, suggesting that glucuronidation starts to become the preferred route. After this point, glucuronidation expression starts decreasing, and both NAPQI formation and sulphation increase. Research shows that at toxic doses of paracetamol, the sulphation and glucuronidation pathways both become saturated, resulting in large amounts of paracetamol undergoing the toxic, NAPQI-forming pathway (Rotundo & Pyrsopoulos, 2020). At overdose levels of paracetamol, however, this is not the case, with active CYP2E1 enzymes being killed, and their expression levels being downregulated. This is most likely the behaviour seen in this case, with the increase in expression at 4 days of treatment showing partial recovery behaviour (AlWahsh et al., 2019).

2.2. GSH de novo synthesis

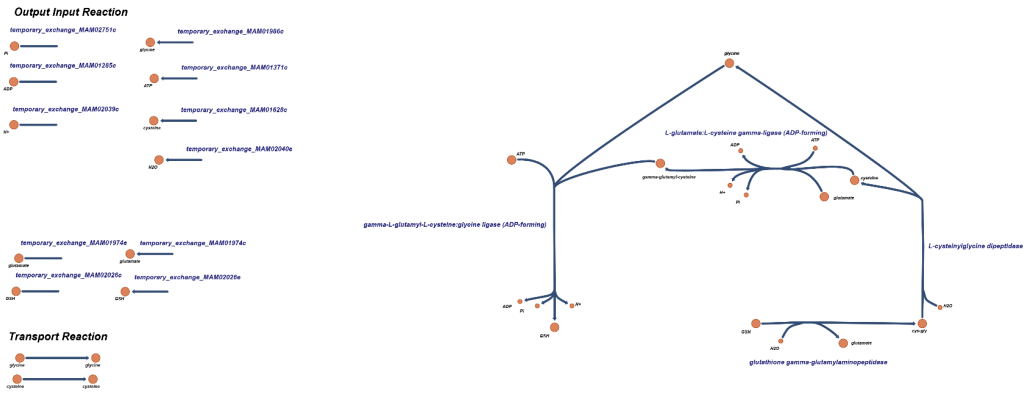


Figure 2. *Expanded GSH de novo synthesis*

There was an existing GSH de novo synthesis task in the model, which was modified and expanded to include its biologically cyclic nature. The final mapped version can be found in Figure 2. GSH starts in the extracellular space, and through the metabolic reactions seen above, it gets reformed in the cytoplasm.

The first reaction uses up GSH and transforms it into cysteine-glycine and is mediated by the gamma-glutamyltransferase (GGT) family of enzymes. After the cell is treated with paracetamol, the gene expression of these enzymes increases, peaking at Sample 3 (3 hours of treatment with paracetamol), after which it starts decreasing. Its elevated levels in the bloodstream are often used as a biomarker for liver damage (Weber et al., 2022). This is because, as GSH reserves are depleted by NAPQI, the liver cells start to undergo damage and GGTs are released into the bloodstream (Gamma-Glutamyl Transferase (GGT) Test, n.d.). The initial increase in GGT expression might be caused by the cell's compensatory reaction, attempting to replenish the lost enzymes. After 3 hours of exposure, the cell starts to undergo significant liver damage, and therefore, this compensatory reaction is stopped. This may be an indicator of the exact point at which liver damage begins to be significant, after 3 hours of treatment with paracetamol.

By looking further into the task, the cysteine and glycine forming reaction is catalysed by aminopeptidases, which hydrolyse the bond in cysteine-glycine, to form cysteine and glycine. The healthy sample has the lowest expression of this enzyme compared to the paracetamol-treated samples. Its gene expression peaks at 4 days of treatment with paracetamol. Typically, an increased aminopeptidase level in serum (Rutenburg et al., 1964) is an indicator of liver damage, and, as with the GGT levels, the increase in the enzyme's expression is likely a compensatory mechanism, where the cell is attempting to replenish the lost reserves. Increased aminopeptidase levels are expected for cells in the recovery stages of cell damage (Nwokolo, 2025).

What is interesting to note about the increase in expression levels of this enzyme is that it is not linear, with a decrease being shown after 3 hours (Sample 3), which may be the result of less GSH being available to be metabolised in the beginning stages of NAPQI action. As damage to hepatocytes begins, after 12 hours (Sample 5), and the enzymes are released from the cells, the compensatory increase in expression is seen. The compensatory increase in gene expression could be an explanation for the behaviour exhibited in this case, but the more likely explanation for this increase is the cell's recovery mechanism.

Cysteine is then turned into gamma-glutamyl-cysteine by glutamate-cysteine ligases (GCLs). The gene expression GCL is kept about constant until Sample 7 (4 days exposure to paracetamol), when it drops drastically. This would suggest that the action of this enzyme's expression is not impacted by paracetamol exposure, but is impacted by recovery. It was found that cells with overexpressed GCL are more resistant to apoptosis (Botta et al., 2004) and play an important adaptive role in the recovery from injury (Krejsa et al., 2010). This further supports the hypothesis that Sample 7 shows cells in the recovery stage.

Gamma-glutamyl-cysteine and glutamate are catalysed by glutathione synthetase (GSS) to synthesise cytosolic GSH in the final step of the pathway. The expression of this enzyme peaks in Sample 5 (12 hours of treatment with paracetamol), and decreases afterwards.
This might be because, as NAPQI starts to cause liver damage (after 3 hours as shown above), the neutralisation mechanism is activated, which involves producing more GSH through increasing the expression of GSS (Spielberg & Gordon, n.d.).

This task and its related gene expression give insights into the different stages of paracetamol action and how they affect gene expression levels of GSH metabolism. It can be concluded that significant liver damage starts to occur at 3 hours of treatment with paracetamol, at which point the cell releases GGTs into the bloodstream, leading to an increase in this enzyme's gene expression in an attempt to compensate for the lost enzymes. The cell also begins its NAPQI neutralisation mechanism at this point, increasing the gene expression of GSSs. The peak expression of GSS is at 12 hours of treatment with paracetamol, showing that at this point the damage is most severe. This is not in accordance to literature, since it is generally regarded that hepatotoxicity begins between 24 and 48 hours (Agrawal et al., 2025). Expression of aminopeptidases peaks at 3 hours of treatment, after which point it decreases, and starts to increase again after 5 hours. This is most likely due to the release of enzymes into the bloodstream at the point of most severe damage, and the increase in expression is a compensatory mechanism for this loss.

2.3. TCA cycle

Research shows that the formation of NAPQI inhibits the electron transport chain, the aerobic respiration pathway is expected to be altered. The Citric acid cycle was mapped, starting from glycolysis (Appendix B).

In the healthy sample, it is observed that the conversion of 11-cis-retinal to 11-cis-retinol has a very high gene expression, over double that of the rest of the reactions. This reaction is mediated by hydroxysteroid dehydrogenases, retinol dehydrogenases, retinaldehyde binding proteins, dehydrogenase/reductases, or the short-chain dehydrogenase/reductase family. Most commonly, retinol dehydrogenase catalyses this reaction (Waagmeester et al., 2023). The expression of this reaction decreases after initial paracetamol exposure, then increases steadily until Sample 5, after which it starts decreasing, with the lowest value present in Sample 7. Deficiency of this enzyme is known to dampen liver injury (Cui et al., 2019), which explains its decrease in the recovery stage (at 4 days). The peak in Sample 5, shows the point at which liver injury is most potent.

In the healthy sample, dihydroxymandelaldehyde is used for NADH metabolism, but when the cell starts being treated with paracetamol, this NADH metabolism starts involving vanillylmandelic acid (VMA) instead. This metabolite is usually produced from the metabolism of noradrenaline and epinephrine in the liver (*Vanillylmandelic Acid - An Overview*, n.d.). NADH metabolite through VMA is usually seen in psychiatric disorders, so this is definitely not an expected result of paracetamol overdose. This is most likely an artefact of GEM modelling, where the enzyme involved in this reaction is upregulated, which causes the algorithm to choose this reaction as the likely one even when it is not biologically accurate. The most likely explanation is that enzymes from the aldehyde dehydrogenase (ALDH) family are upregulated with paracetamol exposure (Wimborne et al., 2020), and since VMA oxidation is one of the reactions mediated by these enzymes, the algorithm includes them as part of the pathway.

Another interesting finding is that this alternative pathway is only found in samples 2 and 3, samples 4, 6, and 7 show normal metabolism of NADH involving dihydroxymandelaldehyde. Sample 5 is another outlier, and chooses a pathway involving homovanillate. All 3 of these pathways involve the same family of enzymes, thus solidifying the hypothesis that this is merely a limitation of such modelling approaches, where a high gene expression of a certain enzyme leads the model to reorganise flux through reactions not relevant to the tissue type.

This finding creates the basis for further improvements in the working of this model and algorithm, but also validates research proposing the idea of increased gene expression of ALDHs in paracetamol overdose.

2.4. Arachidonic acid metabolism

2.4.1. LOX pathways

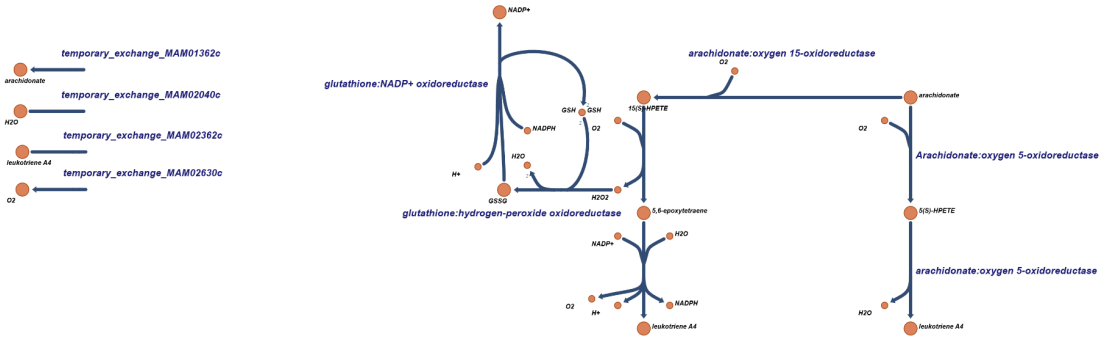


Figure 3. *Arachidonic acid metabolism: The lipooxygenase pathway*

Figure 3 shows the two potential lipooxygenases (LOX) degradation pathways for arachidonic acid. One uses the 5-LOX enzyme to form 5(S)-HPETE as an intermediate before forming leukotriene A4, whereas the other uses 15-LOX to form 15(S)-HPETE as an intermediate.

In the healthy sample, the 5(S)-HPETE pathway is chosen, with a significant gene expression value. For the more diseased samples, the gene expression of the 5-LOX enzyme (Appendix C) increases slightly, but decreases for sample 4, increases for sample 5, then decreases for sample 6. Sample 7, 4 days post continuous treatment with APAP, prefers the 15(S)-HPETE pathway. This pathway also uses GSH to metabolise the H_2O_2 formed in the process. H_2O_2 is a known cause of oxidative stress, and in hepatocytes, it is neutralised using GSH (Li et al., 2000).

The switch from the 5-LOX to 15-LOX pathway with increased stress is not unexpected, with similar behaviour having been observed in cardiovascular disease (Wang et al., 2021). At the beginning of the inflammatory phase, AA is mainly metabolised via the 5(S)-HPETE pathway, since 5-LOX is known to synthesise pro-inflammatory leukotrienes (such as leukotriene A4). In more advanced stages of inflammation, the expression of 15-LOX is increased, redirecting the flux towards the formation of 15(S)-HPETE. Although, unexpectedly, flux would be redirected towards a pathway that requires GSH, when GSH is known to be depleted during paracetamol toxicity, this is merely a limitation of the model and of the algorithm. Metabolite availability is not taken into account when mapping the pathways, only the gene expression.

In hepatocytes, products of 5-LOX (namely 5(S)-HPTE and leukotriene A4) are shown to be precursors to apoptosis and inflammation (Martínez-Clemente et al., n.d.). On the other hand, biologically, the products of 15-LOX include lipoxin A4 (LXA4), which is known for its anti-inflammatory effects. The production of this metabolite is not shown in Figure 3 due to the way in which the task is defined and the logic of the algorithm, but the redirection of flux from 5-LOX towards 15-LOX could be explained by the production of LXA4.

From this task, it can be concluded that until 24 hours of exposure to paracetamol toxicity, the inflammatory/apoptotic pathway is followed, producing 5(S)-HPETE and LTA4. At 4 days post overdose, the expression of 15-LOX is significantly increased, in an attempt to redirect flux from the 5-LOX pathways and produce products with anti-inflammatory properties. This shows that the sample is recovering at this stage.

2.4.2. CYP pathway

Figure 4. *Arachidonic acid metabolism: The cytochrome pathway forming 11,12-DHET*

Figures 4 and 5 show two possible fates of arachidonic acid in the Cytochrome P450 (CYP) pathway: one pathway forming dihydroxyeicosatrienoic acid (DHET), namely 11,12-DHET; and another forming monohydroxyeicosatetraenoic acid (HETE), namely 8-HETE. This pathway is especially interesting to analyse, since the CYP450 enzymes are also involved in the production of the toxic metabolite of paracetamol degradation, NAPQI.

In the DHET pathway (Figure Y), CYP enzymes metabolise arachidonic acid into EETs, while epoxide hydrolases (EPHXs) convert EETs into DHETs. In both healthy and diseased samples, the CYP gene expression was over 10-fold larger than EPHX expression. A potential reason for this could be a problem with the normalisation of the data, since these stark differences are unexpected.

Expression of the CYP enzymes is highest in Sample 5 (12h), followed by the healthy sample. Since the CYP450 enzymes are also involved in the production of paracetamol's toxic metabolite, NAPQI, the elevated expression suggests that it is at this point that most paracetamol is being converted into NAPQI. The second-highest expression was that in the healthy sample, and while it is generally thought that exposure to paracetamol increases the activity of CYP enzymes, some studies found that some isoforms of these enzymes are actually inhibited by high concentrations of paracetamol exposure (Trettin et al., 2014). This would explain the finding, showing that the enzyme's activity is inhibited until 12h of exposure, when it peaks, after which it decreases as the cells begin to undergo recovery.

The expression of EPHXs is nearly constant for samples 1 through 6, but it decreases significantly for sample 7 (4 days of exposure). Increased levels of this enzyme are known to cause apoptosis (Wu et al., 2021), but in this case, the downregulation can be explained by the cell being in recovery.

Figure 5. *Arachidonic acid metabolism: The cytochrome pathway forming 8-HETE*

Figure 5 shows the conversion of arachidonic acid into a monoHETE, specifically 8-HETE. A similar trend to that of the reaction making EETs is observed, which is expected, since this reaction is mediated by the same family of enzymes, CYPs. The highest expression is found in Sample 5, and the second-highest is found in the healthy sample. This further supports the hypothesis that the enzyme is inhibited until 12h of exposure, after which it peaks, which is unexpected in the recovery stage.

3. Critical reflection of the project

My Honour's Research Project focused on understanding the MaCSBio metabolic pathway optimisation algorithm, analysing the results produced in a useful way, and finally, conducting my own research in modelling the metabolic pathways of paracetamol toxicity. The findings back up existing investigations done in this area, showing the usefulness and practicality of in-silico metabolic modelling to better understand drug metabolism and its effect on normal metabolic functioning. This project also aimed to expand and modify the Human1 GEM to include more accurate paracetamol-related reactions, which was achieved. While there is certainly more research to be done on this topic, I believe this project produced useful and interesting findings in the metabolism of paracetamol, its toxic effects on the metabolic functioning of the body, as well as pointing out the limitations of constraint-based modelling.

4. References

Acute liver failure. (n.d.). Cedars-Sinai. https://www.cedars-sinai.org/health-library/diseases-and-conditions/a/acute-liver-failure.html

Agrawal, S., Murray, B. P., & Khazaeni, B. (2025a). Acetaminophen toxicity. StatPearls.

Botta, D., Franklin, C. C., White, C. C., Krejsa, C. M., Dabrowski, M. J., Pierce, R. H., Fausto, N., & Kavanagh, T. J. (2004). Glutamate-cysteine ligase attenuates TNF-induced mitochondrial injury and apoptosis. Free Radical Biology and Medicine, 37(5), 632–642. https://doi.org/10.1016/j.freeradbiomed.2004.05.027

Cui, X., Ma, B., Wang, Y., Chen, Y., Shen, C., Kuang, Y., Fei, J., Lu, L., & Wang, Z. (2019). Rdh13 deficiency weakens carbon tetrachloride-induced liver injury by regulating Spot14 and Cyp2e1 expression levels. Frontiers of Medicine, 13(1), 104–111. https://doi.org/10.1007/s11684-017-0568-x

Forman, H. J., Zhang, H., & Rinna, A. (2009). Glutathione: Overview of its protective roles, measurement, and biosynthesis. Molecular Aspects of Medicine, 30(1–2), 1–12. https://doi.org/10.1016/j.mam.2008.08.006

Gamma-glutamyl transferase (GGT) test. (n.d.). https://medlineplus.gov/lab-tests/gamma-glutamyl-transferase-ggt-test/

Glutamate-cysteine ligase attenuates TNF-induced mitochondrial injury and apoptosis. (n.d.). Free Radical Biology and Medicine, 37(5), 632–642. https://doi.org/10.1016/j.freeradbiomed.2004.05.027

Krejsa, C. M., Franklin, C. C., White, C. C., Ledbetter, J. A., Schieven, G. L., & Kavanagh, T. J. (2010). Rapid Activation of Glutamate Cysteine Ligase following Oxidative Stress. Journal of Biological Chemistry, 285(21), 16116–16124. https://doi.org/10.1074/jbc.m110.116210

Li, J., Chiew, A. L., Isbister, G. K., & Duffull, S. B. (2021). Sulfate conjugation may be the key to hepatotoxicity in paracetamol overdose. British Journal of Clinical Pharmacology, 87(5), 2392–2396. https://doi.org/10.1111/bcp.14642

Li, J., Huang, C. Y., Zheng, R. L., Cui, K. R., & Li, J. F. (2000). Hydrogen peroxide induces apoptosis in human hepatoma cells and alters cell redox status. Cell Biology International, 24(1), 9–23. https://doi.org/10.1006/cbir.1999.0438

Martínez-Clemente, M., Ferré, N., González-Périz, A., López-Parra, M., Horrillo, R., Titos, E., Morán-Salvador, E., Miquel, R., Arroyo, V., Funk, C. D., & Clària, J. (n.d.). 5‐lipoxygenase deficiency reduces hepatic inflammation and tumor necrosis factor α–induced hepatocyte damage in hyperlipidemia‐prone ApoE‐null mice. Hepatology, 51(3), 817–827. https://doi.org/10.1002/hep.23463

NAPQI - An overview. (n.d.). ScienceDirect Topics. https://www.sciencedirect.com/topics/pharmacology-toxicology-and-pharmaceutical-science/napqi

Nwokolo, G. C. (2025). Aminopeptidase N (CD13): Bridging physiology, pathology and therapeutic potential. In Advances in Clinical Chemistry (pp. 207–269). Elsevier. https://doi.org/10.1016/bs.acc.2025.07.002

Rotundo, L., & Pyrsopoulos, N. (2020). Liver injury induced by paracetamol and challenges associated with intentional and unintentional use. World Journal of Hepatology, 12(4), 125–136. https://doi.org/10.4254/wjh.v12.i4.125

Rutenburg, A. M., Banks, B. M., Pineda, E. P., & Goldbarg, J. A. (1964). A comparison of serum aminopeptidase and alkaline phosphatase in the detection of hepatobiliary disease in anicteric patients. Annals of Internal Medicine, 61(1), 50–55. https://doi.org/10.7326/0003-4819-61-1-50

Spielberg, S. P., & Gordon, G. B. (n.d.). Glutathione synthetase–deficient lymphocytes and acetaminophen toxicity. Clinical Pharmacology & Therapeutics, 29(1), 51–55. https://doi.org/10.1038/clpt.1981.9

Trettin, A., Böhmer, A., Suchy, M.-T., Probst, I., Staerk, U., Stichtenoth, D. O., Frölich, J. C., & Tsikas, D. (2014). Effects of paracetamol on NOS, COX, and CYP activity and on oxidative stress in healthy male subjects, rat hepatocytes, and recombinant NOS. Oxidative Medicine and Cellular Longevity, 2014, 1–12. https://doi.org/10.1155/2014/212576

Vanillylmandelic Acid - An overview. (n.d.). ScienceDirect Topics. https://www.sciencedirect.com/topics/neuroscience/vanillylmandelic-acid

Wang, B., Wu, L., Chen, J., Dong, L., Chen, C., Wen, Z., Hu, J., Fleming, I., & Wang, D. W. (2021). Metabolism pathways of arachidonic acids: Mechanisms and potential therapeutic targets. Signal Transduction and Targeted Therapy, 6(1), 94-. https://doi.org/10.1038/s41392-020-00443-w

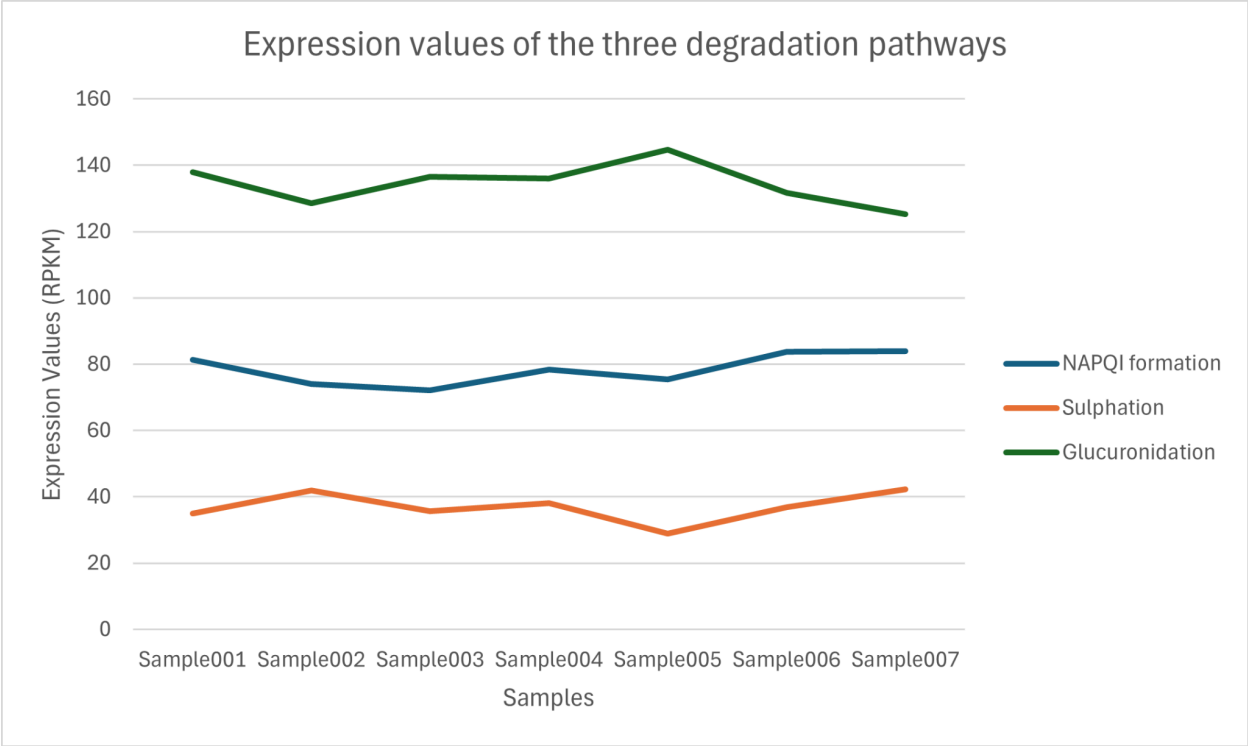
Weber, S., Allgeier, J., Denk, G., & Gerbes, A. L. (2022). Marked increase of gamma-glutamyltransferase as an indicator of drug-induced liver injury in patients without conventional diagnostic criteria of acute liver injury. Visceral Medicine, 38(3), 223–228. https://doi.org/10.1159/000519752

Wimborne, H. J., Hu, J., Takemoto, K., Nguyen, N. T., Jaeschke, H., Lemasters, J. J., & Zhong, Z. (2020). Aldehyde dehydrogenase-2 activation decreases acetaminophen hepatotoxicity by prevention of mitochondrial depolarization. Toxicology and Applied Pharmacology, 396, 114982. https://doi.org/10.1016/j.taap.2020.114982

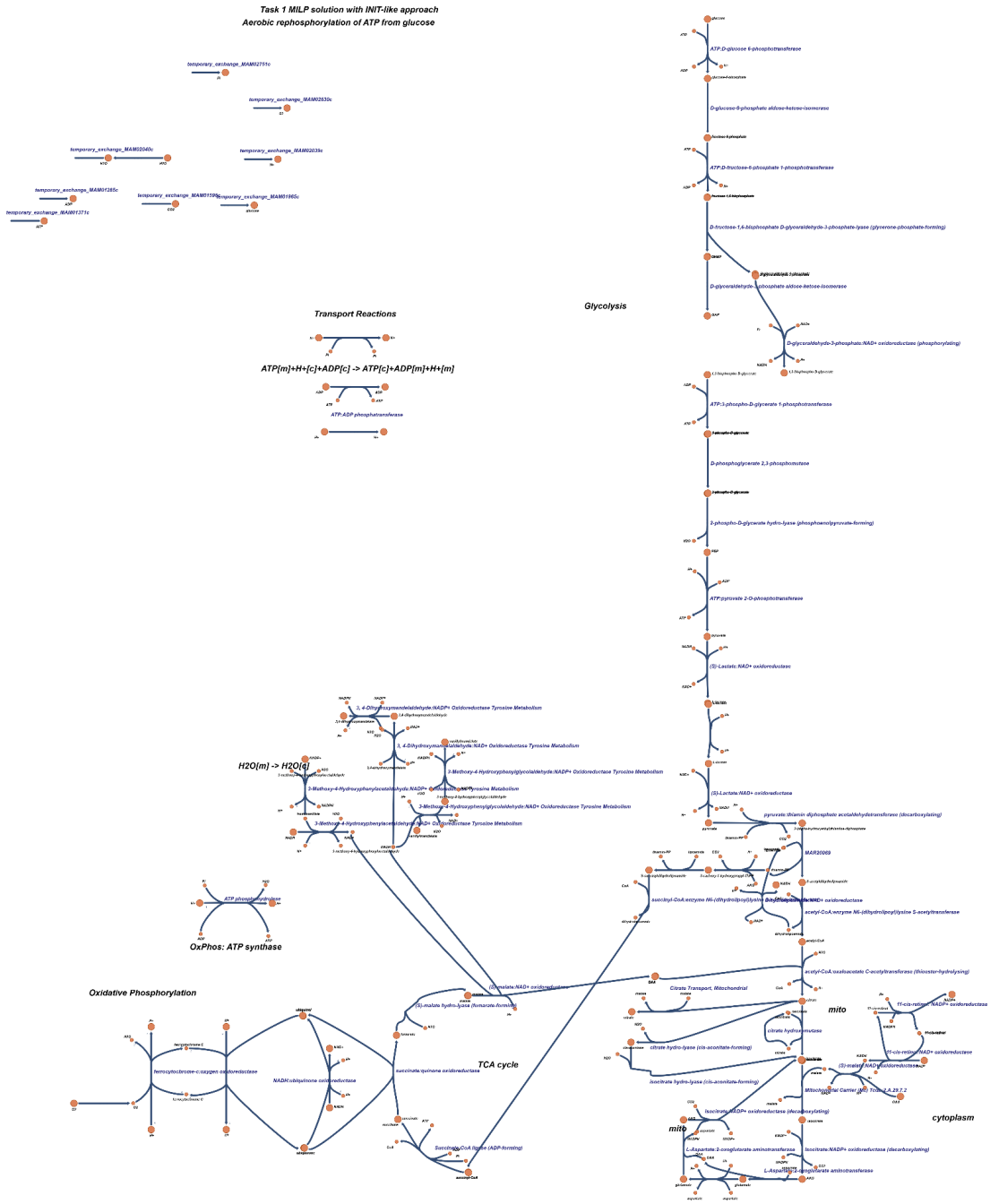
Wu, J., Fan, Z., Zhao, Y., Chen, Q., & Xiao, Q. (2021). Inhibition of soluble epoxide hydrolase (sEH) protects hippocampal neurons and reduces cognitive decline in type 2 diabetic mice. The European Journal of Neuroscience, 53(8), 2532–2540. https://doi.org/10.1111/ejn.15150

5. Appendix

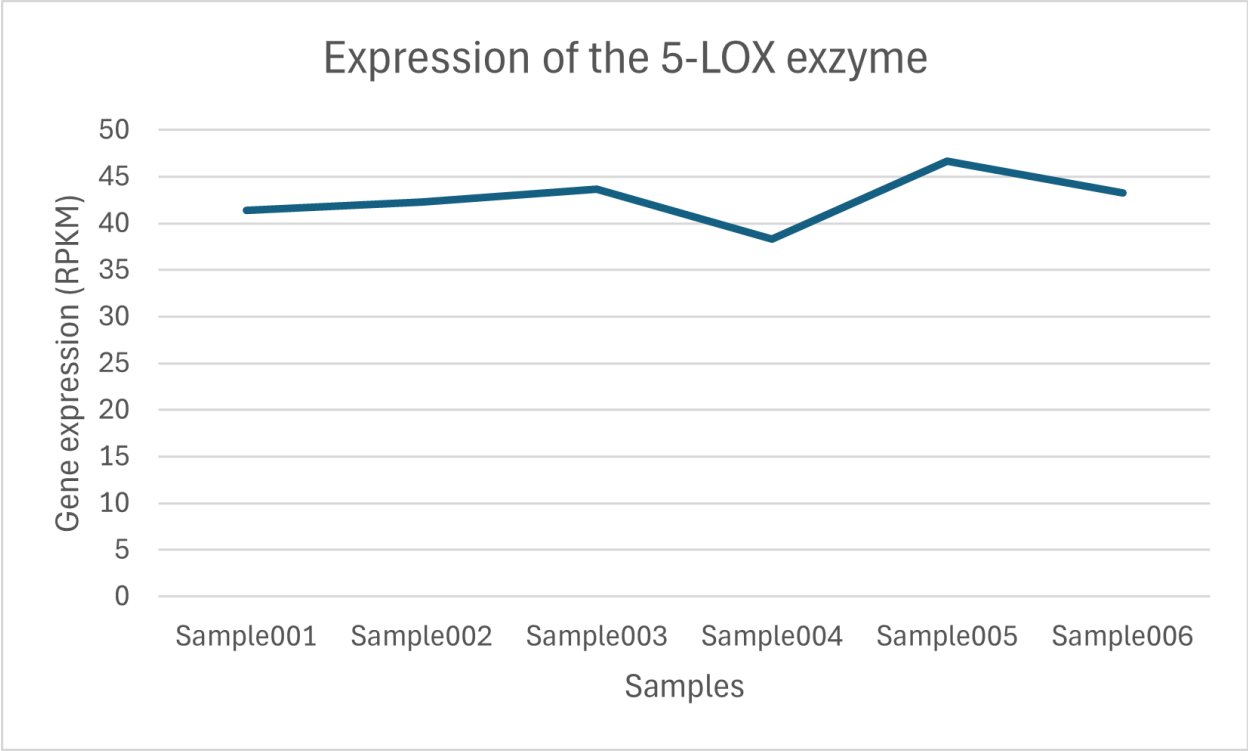
A.


Expression values of the three degradation pathways
Expression Values (RPKM)
160
140
120
100
80
60
40
20
0
Sample001
Sample002
Sample003
Sample004
Sample005
Sample006
Sample007
Samples
NAPQI formation
Sulphation
Glucuronidation

B.


Task 1 MILP solution with INIT-like approach
Aerobic rephosphorylation of ATP from glucose
Glycolysis
Transport Reactions
ATP[m]+H+[c]+ADP[c] -> ATP[c]+ADP[m]+H+[m]
ATP:ADP phosphotransferase
H2O[m] -> H2O[c]
OxPhos: ATP synthase
Oxidative Phosphorylation
TCA cycle
mito
cytoplasm

C.


Expression of the 5-LOX exzyme
Gene expression (RPKM)
50
45
40
35
30
25
20
15
10
5
0
Sample001
Sample002
Sample003
Sample004
Sample005
Sample006
Samples